

Ocean3D | SIH26067 completion audit

Evidence reviewed on 8 September 2026. Prepared for the project team and technical reviewers.

Verdict: core workflow implemented; complete delivery unproven

No, we cannot honestly call the problem completely solved yet. Ocean3D implements much of the requested browser workflow for supported datasets. It co-displays real model fields and instrument observations and exposes working scientific APIs. The remaining issues include a concrete native-grid integration gap, incomplete operational source coverage, and uncompleted scientific, infrastructure and user acceptance.

This assessment uses the full problem statement supplied in this conversation, current code and API responses, earlier recorded tests, and primary-source research. It is not a fresh exhaustive test run or independent certification.

Requirement coverage

Requirement	What is implemented	Completion boundary
Browser-native 3D	React and vtk.js/WebGL volumes, full-depth slices, isosurfaces, sections, time animation and current vectors	Regional rectilinear grids; source and display resolution are bounded
Unified instrument view	Argo, glider, CTD and BGC markers; clickable depth charts, timestamps, QC and source information	Real historical cases; current institutional mission feeds are not integrated
User controls	Variables, palettes, min/max, linear/log scale, independent layer opacity and vertical exaggeration	Implemented; target-device performance and broad accessibility remain to be evaluated
Automated ingestion	xarray NetCDF and delimited text; metadata validation, source adapters and persistent import manifests	Supported contracts, not arbitrary NetCDF; native sigma/curvilinear model grids need adaptation
Extensible design	Registered sensors, variables and derived products; mooring, ADCP and HF-radar contracts	Vendor decoders still need adapters; ML output support is not a trained model
Scalable web service	FastAPI REST, bounded caches, lazy archive reads and immutable multi-worker mode	Local checks exist; representative INCOIS deployment and sustained workload are unverified
Interoperability	WMS 1.3.0, WCS 1.0.0 and CF-aware import/export within documented subsets	External interoperability acceptance remains pending; no blanket standards-conformance claim
Outreach	Four interactive lessons connected to the explorer	Implemented; no student or educator evaluation yet

Evidence that the core is real

Fresh read-only probes returned a ready API with **14 registered models and 29 observation records**, zero restoration errors, a valid binary current-volume response, and WMS/WCS capability documents. Counts include synthetic fixtures and retained source versions, not 14 independent operational feeds. A historical HYCOM/Argo comparison reproduced **1,001 matched levels and 0.65527 C temperature RMSE**; one cast with unknown assimilation independence does not establish general forecast skill.

What prevents complete delivery

1. Successful polling does not establish current ocean coverage

The public INCOIS ERDDAP metadata advertises 16 scientific datasets. Its four depth-resolved Argo analyses have **1-degree spacing**; the 10-day products end on **30 July 2026**, monthly products on **15 July 2026**, and the Indian Argo table on **23 April 2025**. These are source-coverage dates checked directly on 8 September, not download dates. [INCOIS public coverage metadata](#).

Ocean3D's six-hour scheduler is enabled and its retained status reports all 16 products refreshed. It makes bounded selections and preserves older valid generations on failure. It is not a mirror of every archive or every INCOIS service. A separate real HYCOM current case retains three valid times on 7 September 2026, all 40 source depths to 5,000 m, and horizontal stride 12. Those currents are not continuously synchronized.

2. Native operational model geometry is a material integration gap

INCOIS describes ROMS physical-biogeochemical models with approximately **2.25 km coastal / 9 km basin grids and 40 terrain-following s-coordinate levels**. [INCOIS ecosystem model specification](#).

The current generic model normalizer in `backend/science.py:65` requires four independent one-dimensional coordinates and physical depth in metres or kilometres. It explicitly rejects curvilinear and sigma grids. **Inference:** raw native terrain-following output cannot be assumed to load correctly. Obtain representative files and implement a verified adapter or conversion. An already regridded physical-depth export may fit the existing parser; the delivered INCOIS file contract remains unknown. CF defines the metadata needed to reconstruct parametric vertical coordinates. [CF vertical-coordinate conventions](#).

3. Observational and variable coverage is not complete

The live catalog contains real BGC profiles and surface chlorophyll products, but no real three-dimensional chlorophyll model. The synchronized public Argo table lacks data-mode, position-QC and adjusted-error fields needed for richer automated selection. The app explicitly labels its raw-channel choice; the separate GDAC adapter supports raw/adjusted policies. Missing source metadata must remain missing. [INCOIS Argo schema](#), [Argo data-selection guidance](#).

Real glider, CTD and BGC examples demonstrate ingestion and plotting, but not synchronized institutional mission coverage. Some observations cannot be compared with a selected model because time, geographic support or temperature definition differs. Their charts correctly remain available without invented model matches.

What evidence would close the gaps?

4. Operational and human benefits remain unvalidated

Earlier recorded checks include 41 frontend scientific tests, type checking, lint, both frontend builds and overlapping backend suites. Browser interaction checks covered the API and standalone previews. These are prior records, not tests repeated in this audit; no additive or clean full-backend-suite count is claimed.

A recorded local HTTP experiment completed **3,134 requests across 1/2/4 clients**, with four-client p95 latency of **50.47 ms**. Its three small synthetic frames and roughly 30-second total duration do not establish native-archive, GPU or sustained operational capacity. Docker/nginx execution and target institutional identity, TLS and storage remain unverified. The timed-task protocol exists, but **no actual forecaster evaluation has been conducted**, so improved decision speed or accuracy is not established.

Prioritized completion evidence

Priority	Work needed	Evidence required before closing
1 - Source contract	Agree representative INCOIS model and instrument files, access, geometry, variable definitions and expected update cadence	Real sample inventory covering temperature, salinity, currents and the agreed BGC variables; geographic and time overlap
2 - Native ingestion	Adapt terrain-following/curvilinear data where present; preserve coordinates, masks, units and provenance	Independent source-to-display value checks, correct depth reconstruction and any vector rotation/stagger handling required by the supplied model
3 - Complete workflow	Exercise acquisition, rendering, profile inspection, matching, animation, export and restart on representative data	Reproducible cases; correct freshness and QC; incompatible comparisons rejected; stale or failed updates retain truthful status
4 - Institutional acceptance	Test the real deployment, representative concurrent use, independent standards clients and practitioners' tasks	Agreed performance targets on target hardware; successful recovery; scored comparison with the usual workflow; documented limitations

These are proposed acceptance criteria, not invented SIH mandates. Obtain the organizer's or INCOIS team's actual acceptance criteria before claiming full delivery.

Scope and novelty: keep claims defensible

The current React, vtk.js and FastAPI/xarray stack can satisfy the functional intent; vtk.js provides browser WebGL scalar, vector and volumetric methods. There is no demonstrated need to rewrite merely to use a different rendering-library name. [Kitware vtk.js documentation](#).

The statement calls for visualization that supports decisions, not a new forecast engine. A SAR predictor, trained ML system, spherical globe, every CF geometry, OPeNDAP server alongside REST, or newest OGC version is not automatically required. Historical data are valid demonstration inputs: the supplied Copernicus link describes a reanalysis product. [Copernicus GLORYS12V1 description](#).

Browser ocean visualization already has prior art, including MBARI's STOQS. That does not prove it solves the same INCOIS workflow, but a world-first browser-3D claim would be unjustified. The defensible contribution is our integration of model-observation evidence, visible source support and QC, reproducible investigations, and public-source synchronization. A measured forecaster benefit would strengthen that contribution. [MBARI STOQS](#).

Audit trail and source notes

How the assessment was made

The audit compared the user-supplied requirements with renderer, ingestion, synchronization and standards code; inspected source contracts and existing validation records; and made fresh read-only HTTP probes through the running application. It independently retrieved public INCOIS catalog and Argo-schema JSON and reviewed official INCOIS, Argo, CF, OGC, Copernicus, Kitware and MBARI sources. No application code, running service or deployed site was changed for this assessment.

The inspected frontend release is recorded as 18fb4f5; the assessment also includes the current backend workspace, which is not represented by that frontend commit alone. Fresh local probes began at 17:32 UTC on 8 September 2026. Metadata findings are a dated snapshot and may change after publication.

Local evidence for reviewers

- web/components/OceanRenderer.tsx and web/lib/isosurface.ts: actual volume, slice and contour implementation.
- backend/science.py, backend/synchronization.py, backend/ogc.py: coordinate limits, source publication and standards routes.
- docs/REQUIREMENTS_MATRIX.md and docs/INGESTION.md: implemented contracts and explicit boundaries.
- docs/VALIDATION_2026-09-08.md: dated prior scientific, browser, build and recovery checks; sections are chronological and later entries supersede earlier scheduler status.
- docs/benchmarks/http-load-2026-09-08.json and docs/benchmarks/scale-2026-09-07.json: bounded load and synthetic lazy-archive measurements.
- docs/FORECASTER_EVALUATION.md: evaluation protocol and explicit absence of practitioner results.

Primary-source provenance

All linked pages and metadata were accessed on **8 September 2026**. INCOIS catalog/schema JSON are live metadata; their coverage end dates are stated separately on page 2. The INCOIS ecosystem, Argo guidance, Kitware overview and MBARI pages do not display a clear publication date. Copernicus identifies its product as reanalysis; the displayed metadata update is 30 November 2023. CF 1.13 is dated 17 December 2025.

The service implementation advertises WMS 1.3.0 and WCS 1.0.0. The assessment distinguishes documented supported operations and local XML checks from independent conformance testing. It does not require WCS 2.x or claim official certification. [OGC WMS specification](#), [OGC WCS version catalog](#).

Limits of this conclusion

No exact SIH26067 publication was independently located on the official SIH site; the supplied statement is the task's requirement source. Public ERDDAP is not an inventory of all INCOIS holdings. LAS could not be validated through the research browser, and restricted operational files were unavailable. Those access limits do not prove that the data are unavailable to INCOIS. Actual mission formats, target hardware and human evaluation remain open.

Research stopped after the material claims were supported by code or fresh metadata plus relevant primary sources. A deeper literature search cannot replace representative operational files and acceptance tests. The justified claim today is a **working, substantially implemented research prototype for supported datasets**, with a concrete route to complete operational delivery.